This repository contains the MATLAB implementation for the Robust Adaptive Learning Model Predictive Control (RALMPC) framework proposed in the paper Robust MPC for uncertain linear systems. The method is designed for linear systems with parametric uncertainties and additive disturbances performing iterative tasks.

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hannes Update readme 7c3cdc7 · 2 months ago

README.md	Update readme	2 months ago
generatePlot.m	MIT License	2 months ago
get_Initalsolution.m	Add the updated code with more c...	2 months ago
get_P_K.m	Add the updated code with more c...	2 months ago
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solve_OS.m	Add the updated code with more c...	2 months ago
solve_RALMPC.m	Add the updated code with more c...	2 months ago
solve_RAMPC.m	Add the updated code with more c...	2 months ago

README

Robust Adaptive Learning Model Predictive Control (RALMPC) for Linear Systems with Parametric Uncertainties

Overview

This repository contains MATLAB scripts for implementing **Robust Adaptive Learning Model Predictive Control (RALMPC)**. The approach is designed to handle **linear systems with parametric uncertainties**, combining **robust model predictive control (MPC)** with **adaptive learning techniques**. The primary focus is on improving control performance under uncertainty by leveraging iterative learning and parameter adaptation.

Features

- **System Definition**: Defines system dynamics with **parametric uncertainties**.
- **Disturbance Modeling**: Configures **bounded disturbance sets**.
- **Cost Function Setup**: Defines quadratic cost functions for state tracking and control effort.
- **State & Input Constraints**: Implements polyhedral state and input constraints.
- **Adaptive Learning**: Initializes a **hypercube parameter set** for iterative parameter estimation.
- **Offline Computation**: Precomputes necessary parameters for **RALMPC** and computes an initial feasible solution.
- **Simulation for Different Horizons**: Runs **RALMPC and RLMPC** (Robust Learning MPC) for different **prediction horizons (N)**.
- **Result Storage**: Saves computed solutions for further analysis.

Steps for Execution

1. **Perform Offline Computation**
   - Compute required parameters for RALMPC.
2. **Compute Initial Feasible Solution**
   - Generate an initial sample set.
3. **Solve Robust Optimal Control Problem**
   - Solve the robust optimal solution using the true system parameters.
4. **Run Robust Adaptive MPC (RAMPC)**
   - Initialize and solve **RAMPC** with adaptive parameter estimation.
5. **Run RALMPC and RLMPC for Different Horizons**
   - Iterate over different **prediction horizons** and compare results.
6. **Store Results**
   - Saves results in `.mat` files for further analysis.

Folder Structure

This repository contains the following files:

- **`main.m`** : The main script that coordinates the execution.
- **`offlineComputation.m`** : Performs the offline computation of necessary parameters for RALMPC.
- **`get_Initalsolution.m`** : Generates an initial feasible solution for the RALMPC.
- **`solve_OS.m`** : Solves the robust optimal control problem using the true system parameters.
- **`solve_RAMPC.m`** : Solves the Robust Adaptive Model Predictive Control (RAMPC) problem with adaptive parameter estimation.
- **`solve_RALMPC.m`** : Solves the Robust Adaptive Learning Model Predictive Control (RALMPC) problem.

- **`generatedPlot.m`** : Generates plots for visualizing the results of the RALMPC and RAMPC simulations. This includes performance metrics and control trajectories for comparison.

Installation & Requirements

- **MATLAB** (Tested on MATLAB R2023a and later)
- **MPT3 Toolbox** (for polyhedral operations) - Homepage
- **CasADi** (for optimization and nonlinear control) - Homepage

Usage

1. Clone the repository:

```
git clone https://github.com/HannesPetrenz/RALMPC_Linear_Uncertain_Systems.git
cd RALMPC_Linear_Uncertain_Systems
```

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Languages

- MATLAB 100.0%